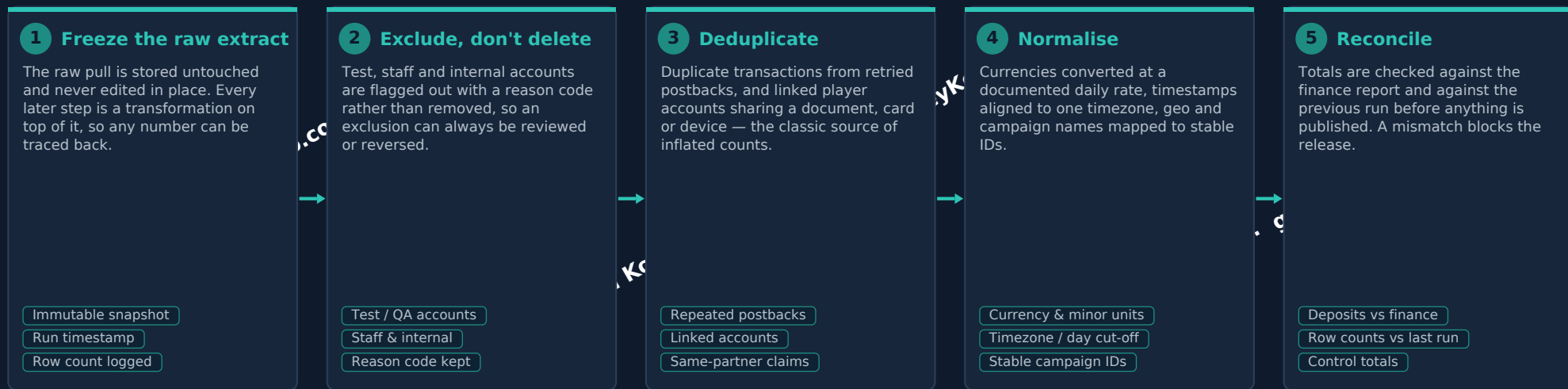


Cleaning & Validation — the step that decides whether the dashboard is true

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Step 3 of the weekly workflow: raw extracts become a trusted dataset, with every exclusion documented and reversible



The judgement call — an extreme value is not automatically an error

In iGaming the biggest numbers are often the real ones. Removing outliers by a blanket rule deletes exactly the players the business cares about most.

€40,000 single deposit Real: VIP player

Kept, flagged for the VIP segment and excluded from average-based charts only.

How it is caught

Cross-checked against the player's KYC tier and deposit history before any decision.

Negative GGR for a day Real: players won

Kept. A losing day for the house is a normal outcome, not a data defect.

How it is caught

Compared with payout ratio for the same day; only investigated if both are extreme.

Bet before registration Error: timezone

Investigated — almost always a timezone or clock-skew issue, fixed at source.

How it is caught

Referential rule: no event may pre-date the player's registration timestamp.

Deposit in cents, not euros Error: minor units

Corrected at source, not patched in the report; the affected period is re-run.

How it is caught

Order-of-magnitude check against the trailing 90-day average deposit.

400 accounts, one device Real, but abuse

Kept and escalated to fraud/RG — a business finding, not a cleaning task.

How it is caught

Device fingerprint and payment instrument grouped; clusters above threshold flagged.

Validation rules and what ships with the numbers

The checks that run every cycle, the thresholds that block a release, and the quality log that goes to stakeholders

Checks that run on every cycle

- **Control totals vs finance**
Deposits, withdrawals and bonus cost matched against the finance report. A gap above tolerance blocks publication until explained.
- **Row counts vs previous run**
A partner file with 40% fewer rows is a delivery failure, not a market trend. Volume is compared before the data is used.
- **Referential integrity**
Every transaction resolves to an existing player, every player to a valid registration. Orphan rows are quarantined, not silently dropped.
- **Impossible values**
Negative stakes, deposits after account closure, activity before registration, future timestamps — each has a rule and an owner.
- **Duplicate transactions**
Same player, same amount, same second — usually a retried postback. Deduplicated on a natural key, not on row order.
- **Linked accounts**
Shared KYC document, payment instrument or device fingerprint. Counted once for player metrics, escalated for fraud review.
- **Currency and minor units**
Order-of-magnitude check against history catches the cents-vs-euros bug before it reaches a revenue chart.
- **Timezone consistency**
All timestamps converted to one reporting timezone with a fixed day boundary, so daily figures reconcile with finance.
- **Late-arriving records**
Chargebacks and settled payments land days later. Periods inside the restatement window are re-run rather than frozen.
- **Bonus vs cash separation**
Bonus-funded activity is tracked but stripped out of revenue, so promotional weeks are not mistaken for organic growth.
- **Self-excluded players**
Accounts under responsible-gambling self-exclusion are handled per policy and never appear in reactivation or marketing segments.
- **Segment coverage**
Every player must fall into exactly one value segment and one geo. Unassigned rows surface as an 'unknown' bucket, never as silent loss.
- **Distribution drift**
Average deposit, bet size and session length compared against the trailing period — a sudden shift usually means a source change, not behaviour.

What I never do silently

- **Overwrite the raw data**
Cleaning happens in a transformation layer. The original extract stays intact and re-runnable.
- **Drop rows without a reason code**
Every excluded record keeps a flag explaining why, so the count of exclusions is itself a metric.
- **Fix a number by hand**
A manual correction that is not in code disappears at the next run and cannot be audited.
- **Change a rule mid-period**
A new exclusion rule is applied to history as well, or the comparison to last week becomes meaningless.
- **Publish through a failed check**
If reconciliation fails, the dashboard ships late with a note — it does not ship wrong and quiet.
- **Reuse a cleaned file blindly**
Cleaning is re-run from raw each cycle. A cached 'clean' extract hides upstream changes that happened since.
- **Round away a difference**
A €200 gap against finance is investigated, not absorbed into rounding. Small unexplained gaps are how big ones start.

Data-quality log — shipped with every dashboard refresh

- Rows in, rows excluded and why, rows out — per source.
- Reconciliation result against finance, with the remaining difference stated in absolute terms.
- Any source that arrived late, partially or not at all during the period.
- Rules changed since the previous run, and whether history was restated.
- A one-line confidence note: what the numbers can and cannot be used for this week.
- Open data issues carried over from previous cycles, with owner and expected fix date.
- Duplicate and linked-account clusters found, passed to fraud review as a separate list.

Output: a trusted dataset plus a written quality log — stakeholders see not only the numbers, but how far they can be trusted